Homogenization benchmark of thin-walled MSG shell models (Reissner–Mindlin and Kirchhoff) against 3D solid finite elements: isotropic and $[45/-45]$ tubes and plate-strips, thin to thick

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Abstract

We benchmark a thin-walled Mechanics-of-Structure-Genome (MSG) shell cross-sectional analysis—in both Reissner–Mindlin (RM, C^0) and Kirchhoff (C^1) wall kinematics—against a 3D solid finite-element cross-sectional analysis (FEniCSx), which here plays the role of VABS. Four sections are studied across a thin-to-thick sweep: a *closed* circular tube and an *open* flat plate-strip, each isotropic and balanced anisotropic $[45/-45]$. The report is results-focused: it reports the full Timoshenko 6×6 stiffness for every case, term by term, with the matrix components labelled by both index and engineering meaning ($C_{11}{=}EA$, $C_{22}{=}GA_2$, $C_{33}{=}GA_3$, $C_{44}{=}GJ$, $C_{55}{=}EI_2$, $C_{66}{=}EI_3$). *Key results.* For the **tube**, both shell models reproduce the solid to $< 1\%$ on every non-zero stiffness at the mid-surface reference across the whole range ($h/R = 0.01$–0.20); the outer-mould-line (OML) reference adds an $O(h/R)$ curvature/arc-length drift. For the **strip**, the axial EA and the strong in-plane bending EI_3 are exact in both models; the open section proved a discriminating test for torsion: it exposed (and we corrected) the RM twist measure, which must carry the full $\kappa_{12}{+}\kappa_{21}{=}{-}2\kappa_1$. With the correction the RM torsion recovers the St-Venant value exactly and *outperforms* Kirchhoff for the thick open section (GJ at $h/W{=}0.20$: RM -0.1% vs. KF $+22\%$). The weak bending EI_2 is captured at the centre (Kirchhoff marginally better; RM slightly soft owing to bend–twist coupling). The transverse shear GA is the genuinely method-sensitive quantity: the thickness-direction GA_3 of a wide strip is a two-dimensional, h^3-governed field that no 1D-shell genome resolves (both models err), and the in-plane GA_2 of the balanced strip is carried by Kirchhoff but under-predicted by the RM C^0 kinematics. The off-diagonal couplings confirm the theory: C_{15}, C_{16}, C_{56} are identically zero at the centroid for these symmetric sections (appearing only as a parallel-axis artifact under an off-centroid reference), while the genuine antisymmetric-$[45/-45]$ couplings are extension–twist (C_{14}) and transverse-shear–bending (C_{25}/C_{36}).

1 Setup

Sections and materials. Tube radius $R = 1$ m, strip width $W = 1$ m; wall/laminate thickness swept over h/R (resp. h/W) $= \{0.01, 0.03, 0.06, 0.12, 0.20\}$. Isotropic: $E = 70\,\text{GPa}$, $\nu = 0.3$

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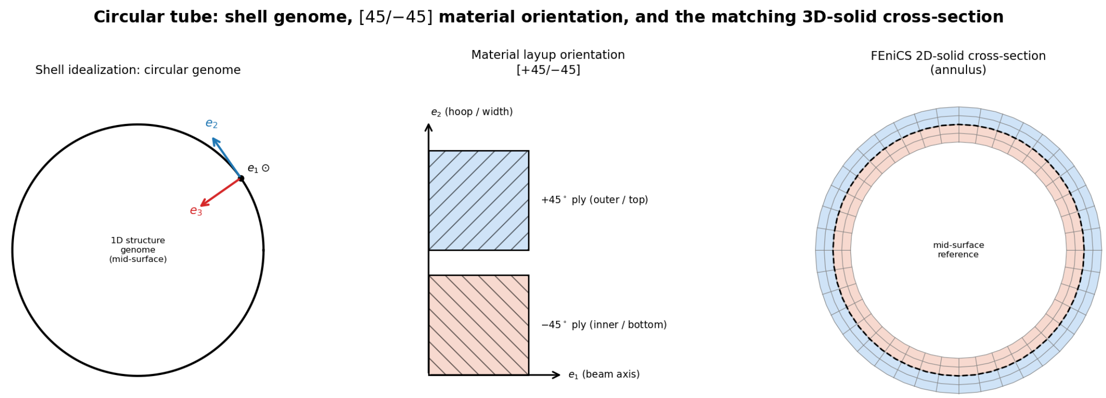

Figure 1: **Tube.** Left: the 1D shell genome (mid-surface circle) with the local frame e_1 (beam, out of plane), e_2 (hoop), e_3 (through-thickness). Centre: the $[+45/-45]$ material orientation (fiber tows shown in each ply, relative to e_1/e_2). Right: the matching FEniCS 2D-solid annulus with the two ply bands and the mid-surface reference.

($G = 26.923\,\text{GPa}$). Anisotropic $[45/-45]$: $E_1 = 37$, $E_2 = E_3 = 9$, $G_{ij} = 4$ GPa, $\nu = 0.3$; the $+45°$ ply is the outer (tube) / top (strip) band and $-45°$ the inner / bottom band. The shell wall is referenced at the mid-surface (*centre*, frac $= 0.5$) or the outer mould line (*OML*, frac $= 0$).

Models. *Solid* — 2D solid cross-sectional analysis (FEniCSx, first-order quadrilaterals, full 3D constitutive, lamina baked into the element frame); $N_c = 200 \times N_r = 16$ (tube) and 120×16 (strip), 8 elements per ply for $[45/-45]$. *Shell* — the 1D MSG structure genome along the wall with the 8×8 plate law $\{\boldsymbol{N}, \boldsymbol{M}, \boldsymbol{Q}\} = [\,\mathbf{A}, \mathbf{B}, \mathbf{0}; \mathbf{B}, \mathbf{D}, \mathbf{0}; \mathbf{0}, \mathbf{0}, \boldsymbol{G}\,]\{\boldsymbol{\varepsilon}^0, \boldsymbol{\kappa}, \boldsymbol{\gamma}^s\}$ condensed to the Timoshenko 6×6, in two kinematics: Kirchhoff (C^1 Hermite, no transverse shear) and Reissner–Mindlin (C^0 Lagrange, transverse shear retained). The geometry, $[45/-45]$ orientation, and the matching solid mesh are shown in Figs. 1–2; the closed tube uses connectivity that wraps, the open strip does not.

2 Results

The study has two independent axes: (i) a *thickness convergence* sweep over $h/R = h/W = 0.01$ to 0.20 (five values) from thin to thick, and (ii) a *reference-surface* comparison (wall mid-surface vs. outer mould line). Every non-zero elastic constant is reported on both axes term by term, at every thickness and both references — **no quantity is averaged or reduced to a mean**; the figures and tables carry the individual values. The MSG plate transverse-shear G (the 8×8 RM block) is used throughout, with no shear-correction factor.

2.1 Circular tube (closed section)

At the **centre** reference the shell is essentially exact on the full 6×6: EA within 0.1% and GA, GJ, EI within $\approx 1\%$ even at $h/R = 0.20$, for *both* RM and Kirchhoff (Fig. 3, 4; Tables 5–10). The balanced $[45/-45]$ tube shows the expected high torsional stiffness ($GJ/EA \approx 0.85$), captured to $< 1\%$. At the **OML** reference the error grows almost linearly in h/R: EA, EI over-predict by $\approx h/2R$ (the $(1 + z/R)$ curvature/arc-length factor) and GA, GJ pick up an $O(h/R)$ drift. The

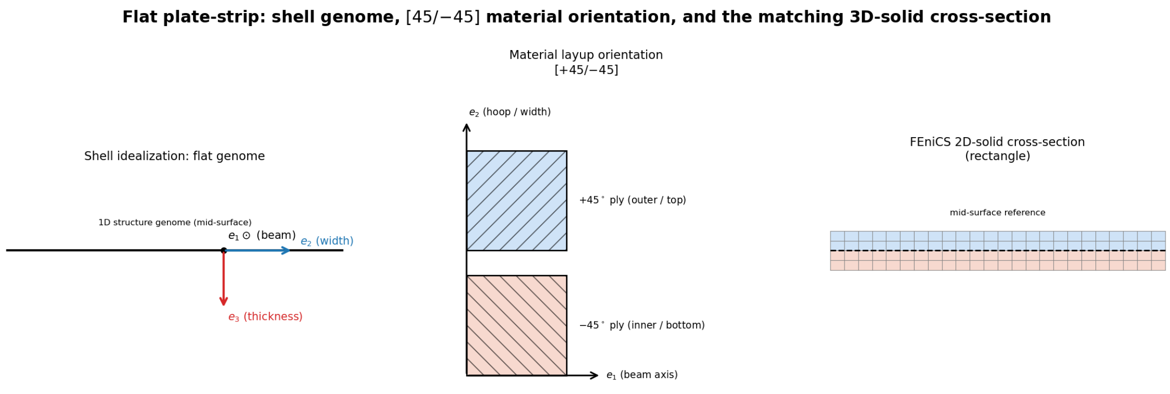

Figure 2: **Plate-strip.** Left: the flat 1D genome with the local frame (e_2 along the width, e_3 through the thickness). Centre: the same $[+45/-45]$ orientation. Right: the matching FEniCS 2D-solid rectangle. The strip is an *open* section (free edges), unlike the closed tube.

off-diagonal couplings are negligible for these symmetric / balanced sections (see the full 6×6, Tables 7, 10).

2.2 Flat plate-strip (open section)

The open strip separates the stiffnesses into three groups (Figs. 5–6; Tables 11–16).

Exact terms — EA and EI_3. The axial stiffness and the strong in-plane bending ($\sim W^3$) match the solid to $< 2\%$ for both models, both materials, across the whole range; the isotropic centre values equal the Euler–Bernoulli closed form ($EA = EWh$, $EI_3 = EhW^3/12$) exactly.

Torsion GJ — the open section corrects and vindicates RM. For a thin open strip the St-Venant constant is $J = Wh^3/3$. The closed tube hides the torsion path because its GJ is membrane (Bredt, A_{66}) dominated; the open strip, having no closed cell, routes all of GJ through the plate-twist measure. This exposed a coefficient error in the RM twist row: the physical plate twist of a beam twisting at rate κ_1 is $\kappa_{12} + \kappa_{21} = -2\kappa_1$, so the macro must carry the full -2; a -1 split (relying on a rotation fluctuation that a thin wall's transverse-shear stiffness locks to ≈ 0) under-counts open-section torsion by exactly a factor of four ($Wh^3/3 \to Wh^3/12$). With the twist measure corrected to -2, the RM torsion is *exact* (GJ within 0.2% at the centre for all h/W) and the closed-tube GJ is preserved ($-1.0\% \to -0.3\%$ at $h/R = 0.20$). Kirchhoff, by contrast, over-stiffens the open-section torsion as the wall thickens (GJ: $+0.5\%$ at $h/W = 0.01$ rising to $+22\%$ at 0.20). The RM model is therefore the more accurate of the two for thick open-section torsion (Fig. 5, panel C_{44}).

Weak bending EI_2 ($\sim h^3$). At the centre this is captured by both models; Kirchhoff is marginally better (within -3 to $+7\%$), while RM runs slightly soft for $[45/-45]$ (-10 to -20%), the expected effect of the bend–twist coupling ($D_{16} \neq 0$) on the free-warping cross-section. The OML reference for EI_2 is dominated by the parallel-axis $EA\,(h/2)^2$ term and is a pure reference offset, not a stiffness error (the OML curves run off-scale in the plots; the values are in the tables).

Transverse shear GA — the method-sensitive group. GA_3 (the thickness-direction shear of a *wide* strip) scales like h^3 and is a genuinely two-dimensional field that a 1D-shell genome cannot represent; both RM and Kirchhoff err on it (-37% thin to $+50$–135% thick), so it is a limit of the shell idealization rather than of either kinematics. GA_2 (the in-plane shear) is exact for the isotropic strip in both models; for the balanced $[45/-45]$ strip Kirchhoff stays within 0–7% while the RM C^0 kinematics under-predict it by ≈ 33–46%. This RM residual is mesh- and quadrature-converged (unchanged under full vs. selectively-reduced integration and under doubling the genome resolution), i.e. it is intrinsic to the present RM open-section shear-warping representation, not a discretization artifact, and marks the open-section transverse-shear solve as the remaining item to refine.

2.3 Isotropic verification against analytic closed forms

For the two isotropic sections every non-zero Timoshenko stiffness has a closed form, which we tabulate as a column beside the FEniCS-solid and the MSG shell (RM, KF; centre) at the thin and thick ends of the sweep (Tables 1–4); the analytic series is also overlaid on the isotropic error plots (Figs. 3, 5, dotted). The transverse shear uses the *MSG plate* transverse-shear stiffness $G = \frac{5}{6}Gh$ (the 8×8 RM plate G block, which carries no shear-correction factor), not an ad-hoc beam coefficient.

Circle (tube). The annulus closed forms $EA = E\pi(r_o^2 - r_i^2)$, $EI = E\frac{\pi}{4}(r_o^4 - r_i^4)$, $GJ = 2EI\,G/E$ match the solid to $< 0.05\%$ at *every* h/R; the transverse shear (the closed-cell wall-membrane value $GA = \frac{1}{2}GA$) is within 1.6% to $h/R = 0.20$. The MSG shell reproduces all of these, confirming the solid is mesh-converged and the shell is exact in the isotropic limit.

Strip. $EA = EWh$ and $EI_3 = EhW^3/12$ are exact; the exact St-Venant rectangular torsion $J = Wh^3[\frac{1}{3} - 0.21\frac{h}{W}(1 - \frac{(h/W)^4}{12})]$ matches to $< 0.15\%$; the in-plane shear $GA_2 = \frac{5}{6}GWh$ (the MSG plate G times the width) matches to $< 0.02\%$. The weak bending $EI_2 = EWh^3/12$ is within -1.9% at $h/W = 0.01$, narrowing as h/W grows — the residual is the finite-width anticlastic (plane-strain) effect the 1D beam/shell omits, not an error. The one term without a usable 1D closed form is GA_3 (thickness-direction shear of a wide strip): the compact $\frac{5}{6}GWh$ over-predicts the true 2D value by orders of magnitude at small h/W (marked †), the same 2D limitation seen in the shell.

2.4 Coupling terms (C_{14}, C_{15}, C_{16}, C_{25}, C_{36}, C_{56})

Figs. 7–9 give the normalized coupling coefficients $|c_{ij}| = |C_{ij}|/\sqrt{C_{ii}C_{jj}}$ (the off-diagonal sign follows each model's axis convention—our solid and shell take opposite through-thickness normals—so the convention-independent comparison is the magnitude; the diagonal terms are sign-independent). Three facts organize the results, and they are consistent with classical lamination theory and the asymptotic thin-walled theory [4, 2, 1]:

- **C_{15} (ext–M_2), C_{16} (ext–M_3), C_{56} (M_2–M_3) are *identically zero* at the centroid** for both the circular and the rectangular section, *regardless of the* $[45/-45]$ *layup*: they require a first moment (C_{15}, C_{16}) or product of inertia (C_{56}) of the axial stiffness about the reference axes, which vanishes for a stiffness distribution symmetric about a centred circle / rectangle. In the data C_{16}, C_{56} sit at the 10^{-13} floor. C_{15} is zero at the centre and jumps to ≈ 0.96 *only* at the OML reference—the textbook parallel-axis (Steiner) artifact of an off-centroid reference line, not a material coupling.

Table 1: Isotropic tube: isotropic Timoshenko stiffness at $h/R = 0.01$ — closed-form **analytic** vs FEniCS-solid vs MSG shell (RM, KF; centre), SI units. The analytic transverse shear uses the MSG plate $G = \frac{5}{6}Gh$.

term	analytic	FEniCS-solid	RM (centre)	KF (centre)	analytic vs solid
C_{11} (EA)	4.3982e+9	4.3975e+9	4.3979e+9	4.3979e+9	+0.02%
C_{22} (GA_2)	8.4581e+8	8.4578e+8	8.4577e+8	8.4566e+8	+0.00%
C_{33} (GA_3)	8.4581e+8	8.4578e+8	8.4577e+8	8.4566e+8	+0.00%
C_{44} (GJ)	1.6917e+9	1.6911e+9	1.6909e+9	1.6909e+9	+0.03%
C_{55} (EI_2)	2.1992e+9	2.1985e+9	2.1985e+9	2.1985e+9	+0.03%
C_{66} (EI_3)	2.1992e+9	2.1985e+9	2.1985e+9	2.1985e+9	+0.03%

Table 2: Isotropic tube: isotropic Timoshenko stiffness at $h/R = 0.20$ — closed-form **analytic** vs FEniCS-solid vs MSG shell (RM, KF; centre), SI units. The analytic transverse shear uses the MSG plate $G = \frac{5}{6}Gh$.

term	analytic	FEniCS-solid	RM (centre)	KF (centre)	analytic vs solid
C_{11} (EA)	8.7965e+10	8.7950e+10	8.7959e+10	8.7988e+10	+0.02%
C_{22} (GA_2)	1.6916e+10	1.7193e+10	1.7063e+10	1.7012e+10	-1.61%
C_{33} (GA_3)	1.6916e+10	1.7193e+10	1.7063e+10	1.7012e+10	-1.61%
C_{44} (GJ)	3.4171e+10	3.4160e+10	3.4071e+10	3.4268e+10	+0.03%
C_{55} (EI_2)	4.4422e+10	4.4408e+10	4.4172e+10	4.4128e+10	+0.03%
C_{66} (EI_3)	4.4422e+10	4.4408e+10	4.4172e+10	4.4130e+10	+0.03%

- **The genuine material couplings of the antisymmetric $[45/-45]$ section are C_{14} (extension–twist, from B_{16}) and C_{25}/C_{36} (transverse-shear–bending).** For the open strip they are large ($|c_{14}| \approx 0.29$, $|c_{25}| \approx 0.43$); the centred shells reproduce C_{14} well (both RM and KF ≈ 0.29), and C_{25} is captured by Kirchhoff (≈ 0.43) but under-predicted by RM (≈ 0.13)—the same open-section shear-warping residual seen in GA_2.
- **The tube's couplings are an order of magnitude smaller** ($|c_{14}|, |c_{25}|, |c_{36}| \approx 0.04$) because the closed wrap-around and the circumferential averaging of the local B-coupling suppress what the flat open strip expresses fully. All couplings vanish for the isotropic sections (Figs. **??**, 9).

Table 3: Isotropic strip: isotropic Timoshenko stiffness at $h/W = 0.01$ — closed-form **analytic** vs FEniCS-solid vs MSG shell (RM, KF; centre), SI units. The analytic transverse shear uses the MSG plate $G = \frac{5}{6}Gh$.

term	analytic	FEniCS-solid	RM (centre)	KF (centre)	analytic vs solid
C_{11} (EA)	7.0000e+8	7.0000e+8	7.0000e+8	7.0000e+8	+0.00%
C_{22} (GA_2)	2.2436e+8	2.2437e+8	2.2437e+8	2.2436e+8	-0.01%
C_{33} (GA_3)	2.2436e+8	8.1825e+5	5.1530e+5	5.0556e+5	†
C_{44} (GJ)	8.9178e+3	8.9290e+3	8.9175e+3	8.9743e+3	-0.13%
C_{55} (EI_2)	5.8333e+3	5.9490e+3	5.8333e+3	5.8333e+3	-1.94%
C_{66} (EI_3)	5.8333e+7	5.8334e+7	5.8334e+7	5.8333e+7	-0.00%

† The compact GA_3 ($= \frac{5}{6}GWh$) over-predicts the wide-strip thickness-direction shear, a 2D plate quantity no 1D model (analytic or shell) captures; only the 3D solid resolves it.

Table 4: Isotropic strip: isotropic Timoshenko stiffness at $h/W = 0.20$ — closed-form **analytic** vs FEniCS-solid vs MSG shell (RM, KF; centre), SI units. The analytic transverse shear uses the MSG plate $G = \frac{5}{6}Gh$.

term	analytic	FEniCS-solid	RM (centre)	KF (centre)	analytic vs solid
C_{11} (EA)	1.4000e+10	1.4000e+10	1.4000e+10	1.4000e+10	+0.00%
C_{22} (GA_2)	4.4872e+9	4.4874e+9	4.4873e+9	4.4872e+9	-0.00%
C_{33} (GA_3)	4.4872e+9	2.5760e+9	6.0518e+9	4.0445e+9	†
C_{44} (GJ)	6.2750e+7	6.2767e+7	6.2708e+7	7.1795e+7	-0.03%
C_{55} (EI_2)	4.6667e+7	4.6692e+7	4.6667e+7	4.6667e+7	-0.05%
C_{66} (EI_3)	1.1667e+9	1.1667e+9	1.1667e+9	1.1667e+9	-0.00%

† The compact GA_3 ($= \frac{5}{6}GWh$) over-predicts the wide-strip thickness-direction shear, a 2D plate quantity no 1D model (analytic or shell) captures; only the 3D solid resolves it.

Table 5: Isotropic tube: principal Timoshenko stiffnesses from the 3D solid (SI units) vs h/R.

h/R	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
0.01	4.3975e+9	8.4578e+8	8.4578e+8	1.6911e+9	2.1985e+9	2.1985e+9
0.03	1.3193e+10	2.5382e+9	2.5382e+9	5.0743e+9	6.5967e+9	6.5967e+9
0.06	2.6385e+10	5.0820e+9	5.0820e+9	1.0156e+10	1.3202e+10	1.3202e+10
0.12	5.2770e+10	1.0209e+10	1.0209e+10	2.0366e+10	2.6476e+10	2.6476e+10
0.20	8.7950e+10	1.7193e+10	1.7193e+10	3.4160e+10	4.4408e+10	4.4408e+10

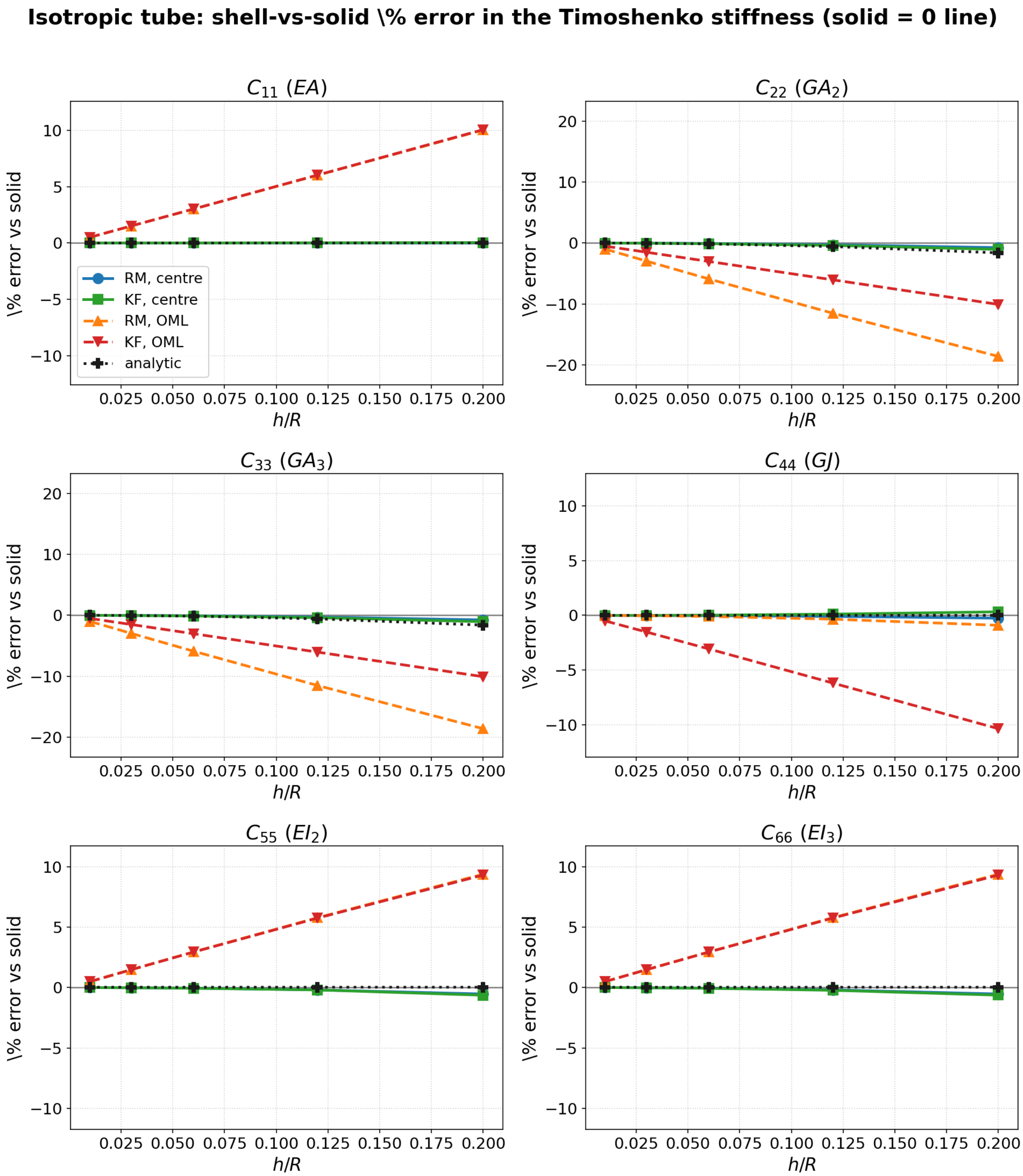

Figure 3: Isotropic tube: % error vs. the solid for each Timoshenko component. Centre-reference curves (solid lines) hug zero through the thick range; OML (dashed) drifts $O(h/R)$; the analytic closed form (dotted, black) overlays the centre curves.

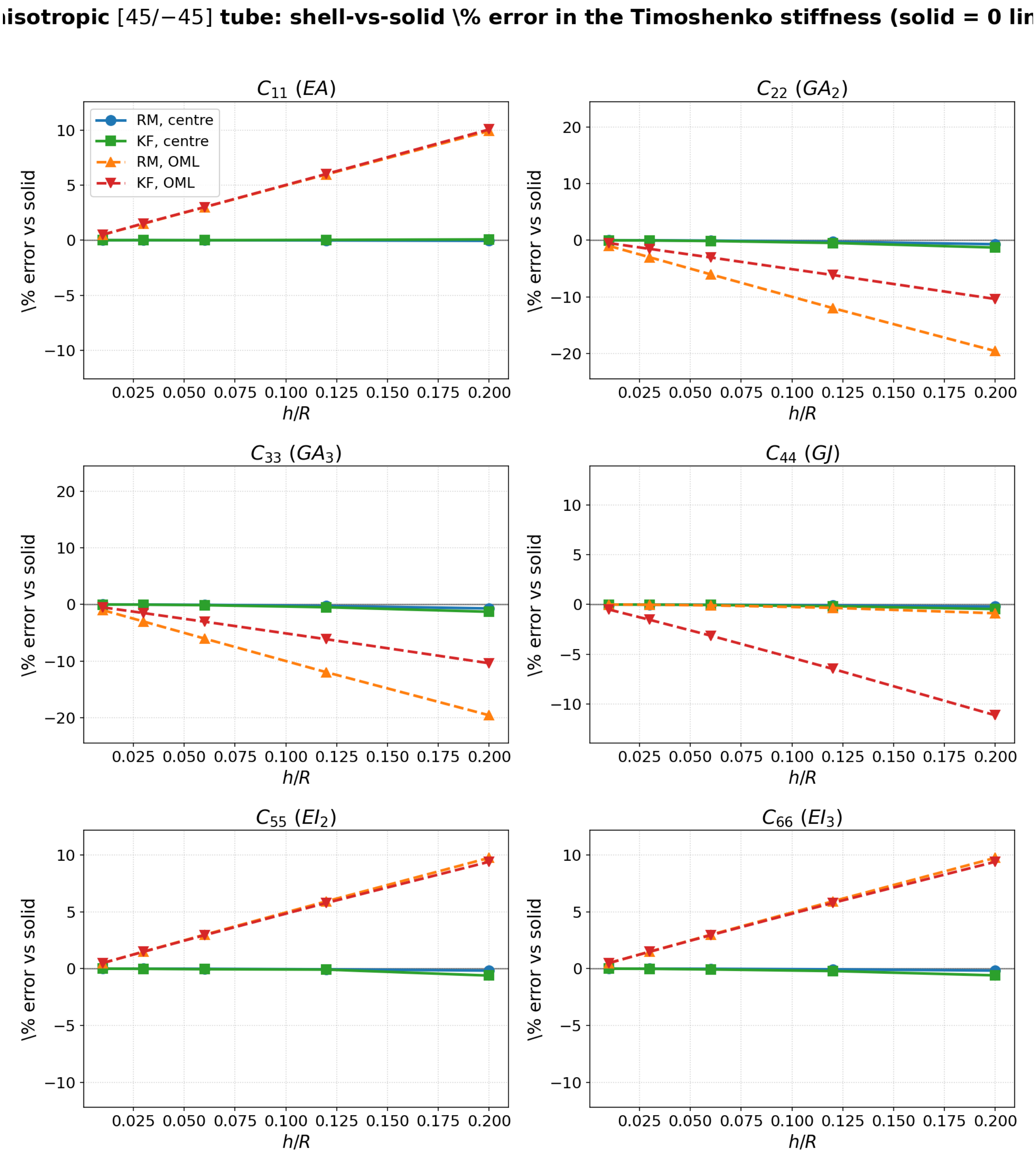

Figure 4: Anisotropic [45/−45] tube: % error vs. the solid. Same behaviour as the isotropic tube; $< 1\%$ at the centre for both models across h/R.

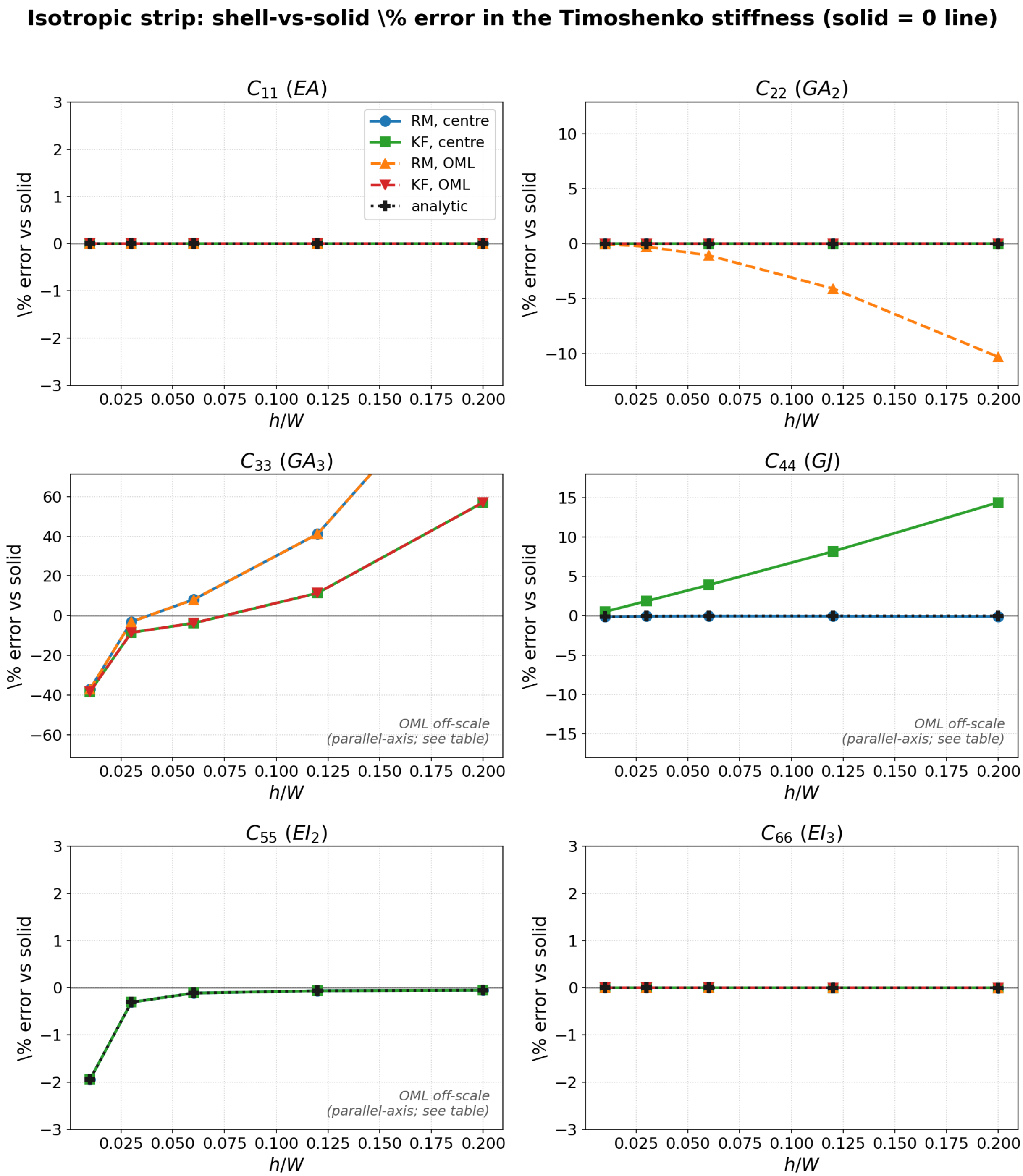

Figure 5: Isotropic strip: % error vs. the solid. EA, EI_3 exact; GJ exact for RM and +0.5–14% for KF; EI_2 within 2% for both; GA_3 off for both (1D-shell limit). OML EI_2/GJ are reference offsets (off-scale). The analytic closed form (dotted, black) overlays the centre curves except for the 2D-limited GA_3.

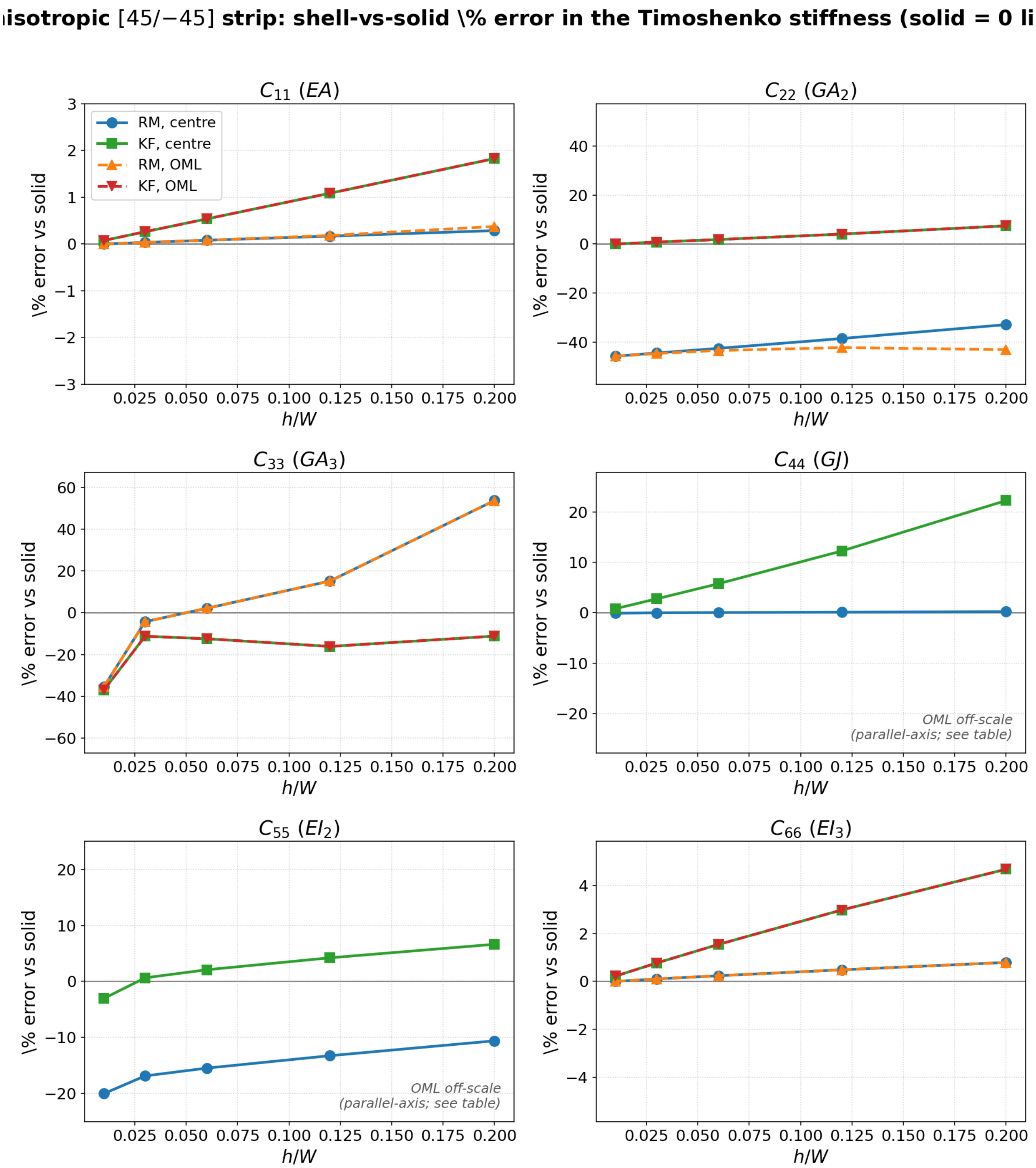

Figure 6: Anisotropic [45/−45] strip: % error vs. the solid. C_{44} (GJ): RM exact, KF rising to +22%. C_{22} (GA_2): KF within 7%, RM $\approx -40\%$ (converged RM open-section residual). C_{33} (GA_3): off for both (1D limit). C_{55} (EI_2): RM soft (bend–twist), KF closer.

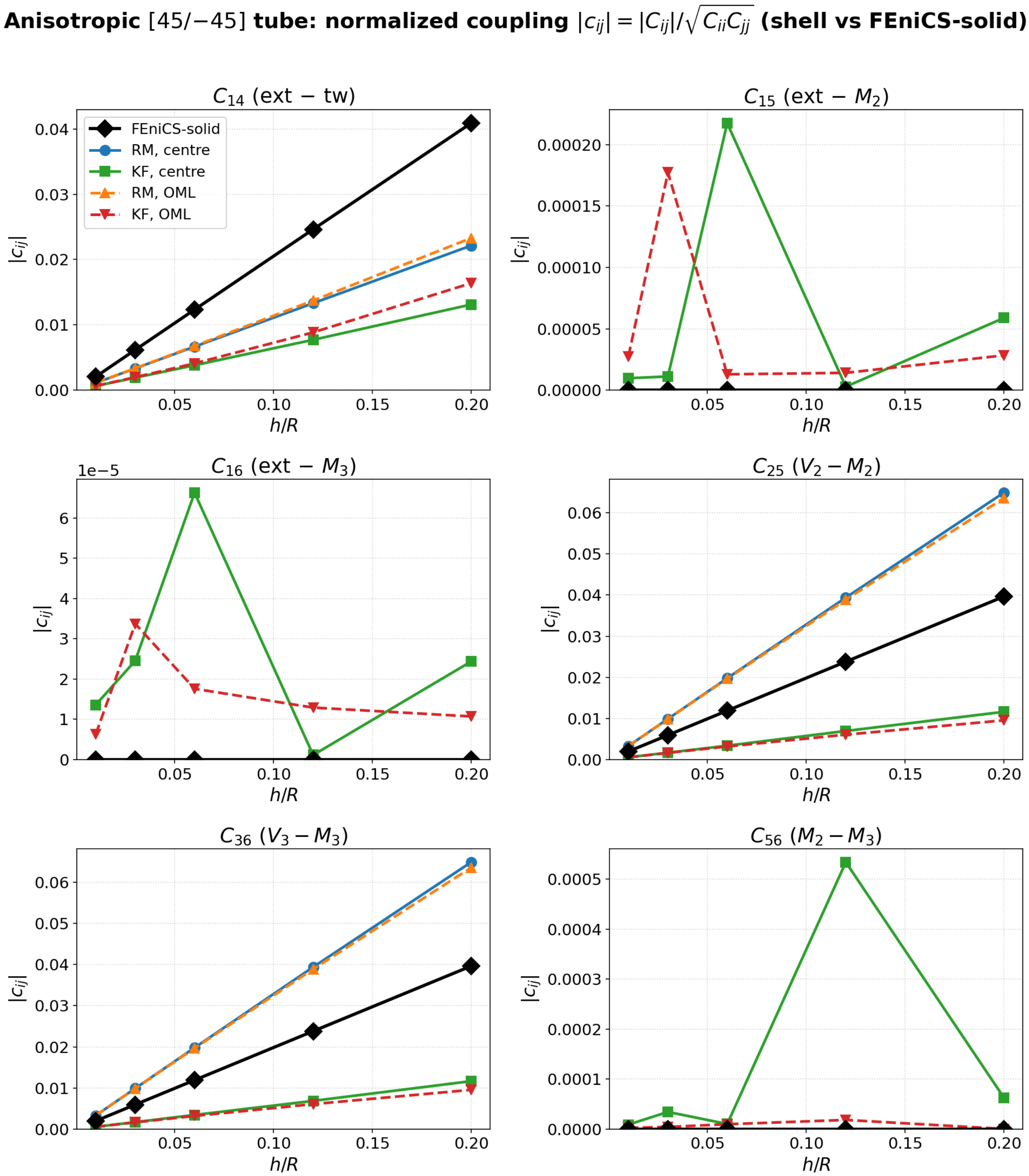

Figure 7: Anisotropic [45/−45] **tube**: normalized couplings vs. the solid. Genuine $C_{14}, C_{25}, C_{36} \approx 0.04$ (suppressed by closure); $C_{15}, C_{16}, C_{56} \approx 0$.

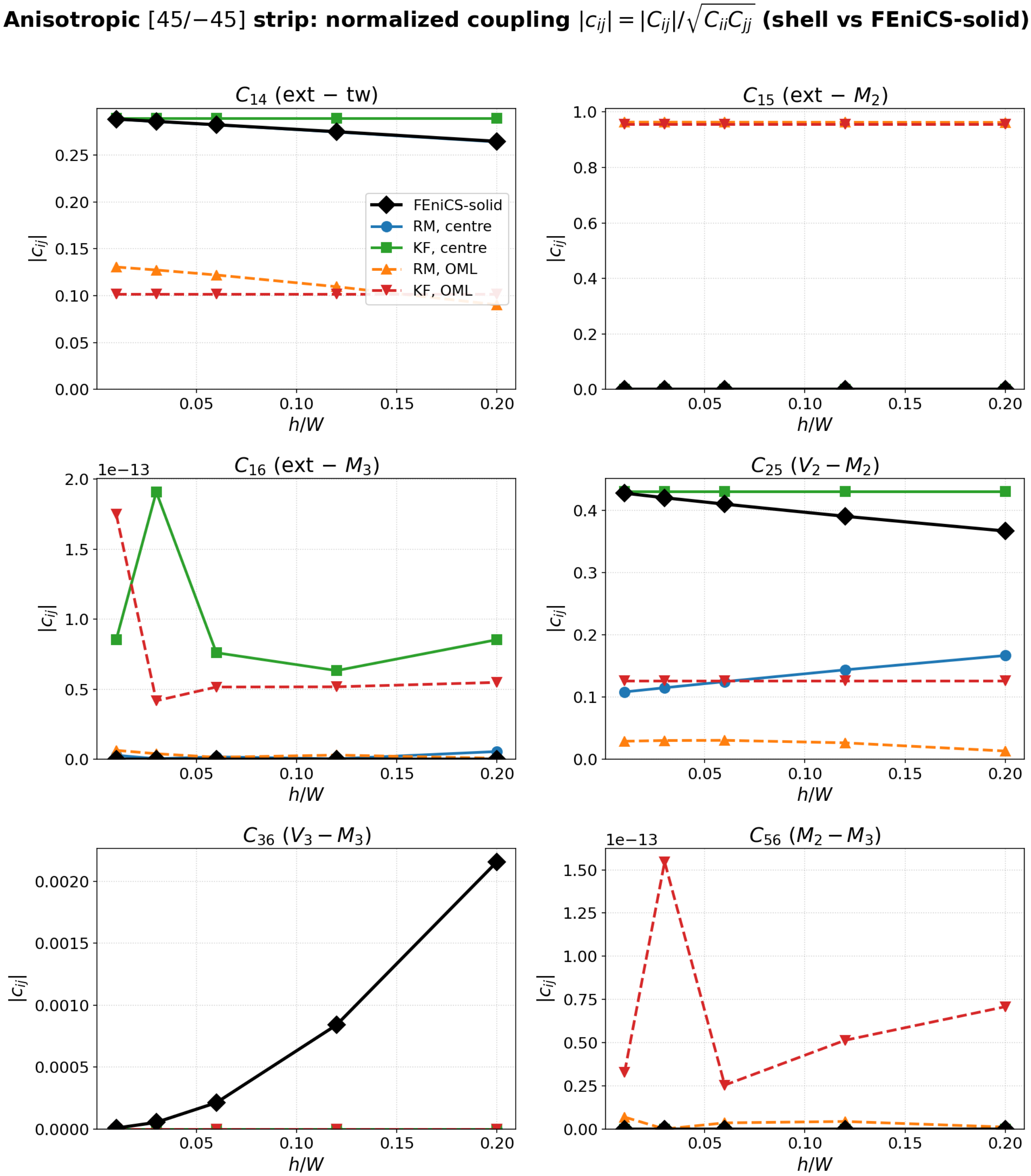

Figure 8: Anisotropic $[45/-45]$ **strip**: C_{14} (≈ 0.29, captured by both models) and C_{25} (≈ 0.43, captured by KF, under-predicted by RM) are the genuine couplings; C_{15} is the OML parallel-axis artifact; $C_{16}, C_{36}, C_{56} \approx 0$.

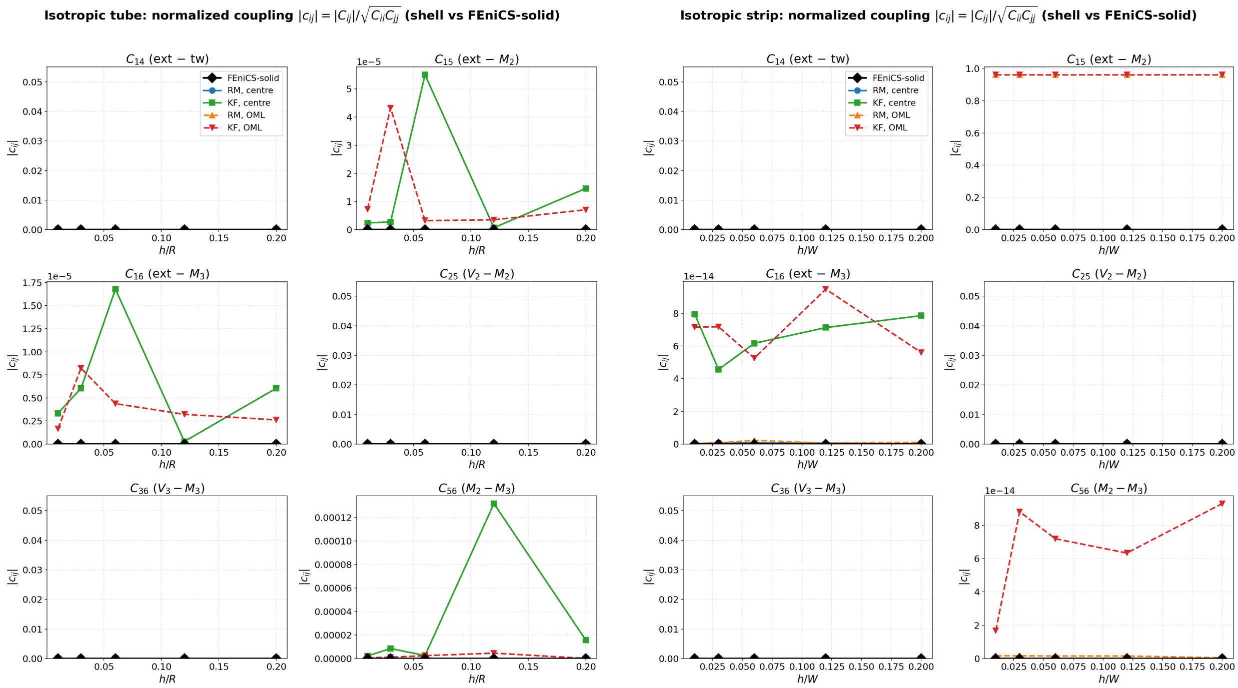

Figure 9: Isotropic tube (left) and strip (right): all material couplings vanish; the only non-zero entry is the C_{15} parallel-axis artifact at the OML reference of the strip.

Table 6: Isotropic tube: shell-vs-solid % error at the thickest section ($h/R = 0.20$).

method	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
RM, centre	+0.01	-0.76	-0.76	-0.26	-0.53	-0.53
KF, centre	+0.04	-1.05	-1.05	+0.32	-0.63	-0.63
RM, OML	+10.01	-18.59	-18.59	-0.91	+9.39	+9.39
KF, OML	+10.05	-10.06	-10.06	-10.35	+9.32	+9.32

Table 7: Isotropic tube: full 6×6 Timoshenko stiffness from the 3D solid at $h/R = 0.20$ (SI). Order [ext, V_2, V_3, tw, M_2, M_3]; off-diagonal couplings are negligible for these symmetric / balanced sections.

8.7950e+10	0.0000e+0	0.0000e+0	0.0000e+0	-0.0000e+0	0.0000e+0
0.0000e+0	1.7193e+10	-0.0000e+0	-0.0000e+0	0.0000e+0	0.0000e+0
0.0000e+0	-0.0000e+0	1.7193e+10	-0.0000e+0	0.0000e+0	0.0000e+0
0.0000e+0	-0.0000e+0	-0.0000e+0	3.4160e+10	0.0000e+0	0.0000e+0
-0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	4.4408e+10	-0.0000e+0
0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	-0.0000e+0	4.4408e+10

Table 8: Anisotropic [45/−45] tube: principal Timoshenko stiffnesses from the 3D solid (SI units) vs h/R.

h/R	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
0.01	7.7054e+8	3.2597e+8	3.2597e+8	6.5178e+8	3.8523e+8	3.8523e+8
0.03	2.3117e+9	9.7812e+8	9.7812e+8	1.9557e+9	1.1559e+9	1.1559e+9
0.06	4.6235e+9	1.9577e+9	1.9577e+9	3.9134e+9	2.3136e+9	2.3136e+9
0.12	9.2485e+9	3.9268e+9	3.9268e+9	7.8437e+9	4.6405e+9	4.6405e+9
0.20	1.5420e+10	6.5890e+9	6.5890e+9	1.3140e+10	7.7873e+9	7.7873e+9

Table 9: Anisotropic [45/−45] tube: shell-vs-solid % error at the thickest section ($h/R = 0.20$).

method	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
RM, centre	-0.05	-0.69	-0.69	-0.22	-0.17	-0.17
KF, centre	+0.08	-1.29	-1.29	-0.46	-0.61	-0.59
RM, OML	+9.94	-19.56	-19.56	-0.88	+9.75	+9.75
KF, OML	+10.08	-10.37	-10.37	-11.13	+9.40	+9.40

Table 10: Anisotropic [45/–45] tube: full 6×6 Timoshenko stiffness from the 3D solid at $h/R = 0.20$ (SI). Order [ext, V_2, V_3, tw, M_2, M_3]; off-diagonal couplings are negligible for these symmetric / balanced sections.

1.5420e+10	0.0000e+0	-0.0000e+0	5.8239e+8	-0.0000e+0	0.0000e+0
0.0000e+0	6.5890e+9	-0.0000e+0	0.0000e+0	-2.8397e+8	-0.0000e+0
-0.0000e+0	-0.0000e+0	6.5890e+9	-0.0000e+0	0.0000e+0	-2.8397e+8
5.8239e+8	0.0000e+0	-0.0000e+0	1.3140e+10	-0.0000e+0	0.0000e+0
-0.0000e+0	-2.8397e+8	0.0000e+0	-0.0000e+0	7.7873e+9	0.0000e+0
0.0000e+0	-0.0000e+0	-2.8397e+8	0.0000e+0	0.0000e+0	7.7873e+9

Table 11: Isotropic strip: principal Timoshenko stiffnesses from the 3D solid (SI units) vs h/W.

h/W	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
0.01	7.0000e+8	2.2437e+8	8.1825e+5	8.9290e+3	5.9490e+3	5.8334e+7
0.03	2.1000e+9	6.7312e+8	1.4911e+7	2.3788e+5	1.5799e+5	1.7500e+8
0.06	4.2000e+9	1.3462e+9	1.1346e+8	1.8659e+6	1.2614e+6	3.5000e+8
0.12	8.4000e+9	2.6925e+9	7.8416e+8	1.4339e+7	1.0086e+7	7.0001e+8
0.20	1.4000e+10	4.4874e+9	2.5760e+9	6.2767e+7	4.6692e+7	1.1667e+9

Table 12: Isotropic strip: shell-vs-solid % error at the thickest section ($h/W = 0.20$).

method	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
RM, centre	-0.00	-0.00	+134.93	-0.09	-0.05	-0.00
KF, centre	-0.00	-0.00	+57.00	+14.38	-0.05	-0.00
RM, OML	-0.00	-10.32	+134.92	+545.06	+1199.31	-0.00
KF, OML	-0.00	-0.00	+57.00	+657.79	+1199.31	-0.00

Table 13: Isotropic strip: full 6×6 Timoshenko stiffness from the 3D solid at $h/W = 0.20$ (SI). Order [ext, V_2, V_3, tw, M_2, M_3]; off-diagonal couplings are negligible for these symmetric / balanced sections.

1.4000e+10	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0
0.0000e+0	4.4874e+9	0.0000e+0	-0.0000e+0	0.0000e+0	0.0000e+0
0.0000e+0	0.0000e+0	2.5760e+9	-0.0000e+0	0.0000e+0	0.0000e+0
0.0000e+0	-0.0000e+0	-0.0000e+0	6.2767e+7	0.0000e+0	0.0000e+0
0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	4.6692e+7	0.0000e+0
0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	-0.0000e+0	1.1667e+9

Table 14: Anisotropic [45/−45] strip: principal Timoshenko stiffnesses from the 3D solid (SI units) vs h/W.

h/W	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
0.01	1.2256e+8	8.5434e+7	4.1047e+4	2.6900e+3	1.1850e+3	1.0198e+7
0.03	3.6700e+8	2.5424e+8	7.8727e+5	7.1254e+4	3.0827e+4	3.0430e+7
0.06	7.3200e+8	5.0340e+8	6.3841e+6	5.5388e+5	2.4313e+5	6.0395e+7
0.12	1.4560e+9	9.8505e+8	5.3317e+7	4.1743e+6	1.9051e+6	1.1910e+8
0.20	2.4091e+9	1.5904e+9	2.3306e+8	1.7743e+7	8.6205e+6	1.9527e+8

Table 15: Anisotropic [45/−45] strip: shell-vs-solid % error at the thickest section ($h/W = 0.20$).

method	C_{11} (EA)	C_{22} (GA_2)	C_{33} (GA_3)	C_{44} (GJ)	C_{55} (EI_2)	C_{66} (EI_3)
RM, centre	+0.29	-32.93	+53.70	+0.22	-10.62	+0.79
KF, centre	+1.83	+7.44	-11.19	+22.28	+6.63	+4.69
RM, OML	+0.38	-43.16	+53.69	+411.81	+1108.15	+0.79
KF, OML	+1.83	+7.44	-11.19	+889.05	+1144.90	+4.69

Table 16: Anisotropic [45/–45] strip: full 6×6 Timoshenko stiffness from the 3D solid at $h/W = 0.20$ (SI). Order [ext, V_2, V_3, tw, M_2, M_3]; off-diagonal couplings are negligible for these symmetric / balanced sections.

2.4091e+9	0.0000e+0	0.0000e+0	-5.4746e+7	0.0000e+0	-0.0000e+0
0.0000e+0	1.5904e+9	0.0000e+0	-0.0000e+0	4.2950e+7	0.0000e+0
0.0000e+0	0.0000e+0	2.3306e+8	0.0000e+0	0.0000e+0	4.6029e+5
-5.4746e+7	-0.0000e+0	0.0000e+0	1.7743e+7	-0.0000e+0	0.0000e+0
0.0000e+0	4.2950e+7	0.0000e+0	-0.0000e+0	8.6205e+6	-0.0000e+0
-0.0000e+0	0.0000e+0	4.6029e+5	0.0000e+0	-0.0000e+0	1.9527e+8

3 Discussion and conclusions

For **closed** thin-walled sections (the tube) the MSG shell is not merely a thin-limit approximation: at the mid-surface reference it reproduces the full 3D solid Timoshenko stiffness to $< 1\%$ well into the thick regime ($h/R = 0.20$), for isotropic and balanced anisotropic walls, in both RM and Kirchhoff forms; the dominant error for thick walls is the OML reference choice (an $O(h/R)$ curvature/arc-length bias), not the wall kinematics. The mid-surface reference is recommended.

The **open** strip is the discriminating case. It cleanly separates terms the shell captures exactly (EA, EI_3, and—with the corrected -2 twist measure—GJ, where RM now matches St-Venant and beats Kirchhoff for thick sections) from terms that are intrinsically harder: the weak bending EI_2 (bend–twist sensitive) and the transverse shear GA. Of the latter, the thickness-direction GA_3 is a fundamental 1D-shell limitation shared by both kinematics, whereas the in-plane GA_2 of the balanced strip is captured by Kirchhoff but under-predicted by the present RM open-section shear-warping representation—a converged, kinematic residual that is the natural target for the next refinement. In short: use the mid-surface reference; trust EA, EI and (with the twist correction) GJ in both models for open and closed sections; and treat the open-section GA as the method-sensitive quantity, with RM's standing advantage being thick-section torsion and the recovery of transverse-shear stress in dehomogenization (documented in the companion paper).

Reproducibility. The FEniCS 2D-solid cases (WSL) are generated by `bench_solid.py` and `bench_strip_solid.py`; the JAX MSG-TW shell cases by `bench_shell.py` and `bench_strip_shell.py`; the tables and figures by `bench_report.py`. Raw 6×6 data are in `data/`. The twist-measure correction is at `rm/msg_rm.py` (`KAPPA12_MACRO` $= -2$).

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